



**DEPARTMENT OF INFORMATION SCIENCE
& ENGINEERING**

A report on

**PlasticDeg – An Intelligent Microbial Plastic Degradation
Potential Prediction System**

submitted to

**Ramaiah Institute of Technology
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submitted by

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Bachelor of Engineering in Information Sciecncce & Engineering

under the guidance of

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CERTIFICATE

This is to certify that the project work titled **PlasticDeg – An Intelligent Microbial Plastic Degradation Potential Prediction System** is a bonafide work carried out by Nishantha HO(1MS23IS083), Roshan Kumar Sharma(1MS23IS104), Saravanan S (1MS23IS112) and Shashwath Prabhu(1MS23IS113) in partial fulfillment for the course of Bachelor of Engineering in Information Science and Engineering during the year 2026. The Project report has been approved as it satisfies the academic requirements with respect to the project work. To the best of our understanding the work submitted in this report has not been submitted anywhere.

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1.

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DECLARATION

We, students of sixth semester BE, Dept. of Information Science and Engineering, Ramaiah Institute of Technology, Bangalore, hereby declare that the project entitled **“PlasticDeg – An Intelligent Microbial Plastic Degradation Potential Prediction System”**, thesis completed and written by us under the guidance of **Dr. Lincy Meera Mathews** Dept. of ISE, Bangalore.

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ABSTRACT

The escalating global crisis of plastic pollution necessitates the development of innovative bioremediation strategies. Microbial plastic degradation, facilitated by specialized enzymes, presents a promising and eco-friendly solution for breaking down synthetic polymers. However, the traditional experimental discovery of novel plastic-degrading enzymes is resource-intensive and time-consuming. This project, "PlasticDeg – An Intelligent Microbial Plastic Degradation Potential Prediction System," introduces a rigorous computational baseline designed to accelerate the discovery of polyester-degrading enzymes. By leveraging advanced protein language models and highly curated biological databases, this system aims to accurately predict plastic-degrading capabilities directly from uncharacterized microbial protein sequences.

The methodology follows a highly structured data processing and machine learning pipeline to ensure robust and honest evaluation metrics. Ground truth positive sequences were meticulously curated from the PAZy database and expanded via BLAST against external libraries under strict 90% identity and query coverage filters. To prevent data leakage and artificially inflated scores, a homology-aware train-test split was executed using MMseqs2, ensuring no protein cluster was divided across the training and testing sets. The predictive architecture utilizes frozen ESM-2 (650M parameter) protein embeddings coupled with downstream classifiers, specifically Random Forest and Logistic Regression algorithms. Furthermore, the training pipeline incorporates a tier-weighted approach to properly balance highly verified "Gold" curated data against homology-propagated "Silver" data.

Evaluation on a held-out homology test set demonstrated exceptional separability, with both classifiers achieving near-perfect Precision-Recall and ROC Area Under the Curve (AUC) scores against broad Swiss-Prot baseline negatives. Specifically, the Random Forest model exhibited a precision of 1.0 with zero false positives at its training-frozen threshold. To validate the model's true robustness and ensure it was not simply identifying generic hydrolases, a rigorous Phase-4 stress test was conducted using reviewed EC 3.1.1 and 3.1.2 Swiss-Prot hydrolases as hard negatives. The system successfully maintained a 0.0 false-positive rate against these confounding enzymes. These results confirm that PlasticDeg successfully leverages deep learning representations to isolate plastic-degrading properties from general protein backgrounds, establishing it as a highly reliable and precise screening pipeline for novel bioremediation candidates.

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1. INTRODUCTION

1.1. General Introduction

The rapid accumulation of plastic waste has become one of the most pressing environmental challenges of the 21st century, necessitating innovative solutions for sustainable degradation. Conventional plastic disposal methods, such as incineration and landfilling, often result in significant ecological footprints and secondary pollution. Consequently, there is an urgent scientific focus on biological degradation pathways, particularly those involving microbial enzymes capable of breaking down complex synthetic polymers.

The PlasticDeg project aims to address this need by developing an intelligent system for predicting the plastic-degradation potential of bacteria and fungi using advanced machine learning techniques. By leveraging state-of-the-art protein language models, specifically the ESM-2 (Evolutionary Scale Modeling) framework, this project seeks to bridge the gap between genomic data and functional enzymatic discovery. The system utilizes a curated database of known plastic-degrading enzymes, primarily sourced from the PAZy database, to train a robust classification pipeline. This pipeline integrates bioinformatics tools like BLAST for dataset expansion and MMseqs2 for homology-aware data splitting to ensure high-quality, non-redundant training. Ultimately, PlasticDeg provides a scalable and technically rigorous baseline for researchers to identify novel biocatalysts, contributing to the global effort of mitigating plastic pollution through biotechnology and intelligent computational screening.

1.2. Problem Statement

Identifying novel plastic-degrading enzymes from vast microbial genomes remains a significant challenge due to the limitations of current computational models. Existing predictive methods frequently suffer from artificially inflated performance metrics caused by sequence redundancy and train-test data leakage, while simultaneously exhibiting high false-positive rates when attempting to distinguish true polyester-degrading enzymes from structurally similar, non-degrading general hydrolases. Consequently, there is a critical need for a rigorous, homology-aware machine learning framework capable of accurately isolating genuine plastic degradation potential from general enzymatic activity, ensuring reliable discovery through strict dataset curation and hard-negative stress testing.

1.3. Objectives of the Project

- **To construct a high-fidelity foundational dataset:** To aggregate and standardize a comprehensive dataset of verified microbial plastic-degrading enzymes, creating a highly reliable ground truth for predictive modeling.
- **To capture deep structural and evolutionary protein features:** To leverage advanced deep-learning protein representations to accurately map the complex biological signatures that dictate plastic degradation potential.
- **To engineer a highly generalizable prediction model:** To design a robust training framework that strictly eliminates sequence redundancy and data leakage, ensuring the model's capacity to discover truly novel, uncharacterized enzymes.
- **To rigorously validate model specificity:** To benchmark the system against confounding general enzymes (hard negatives), ensuring the framework explicitly targets synthetic polymer degradation with an effectively minimized false-positive rate.

1.4. Project Deliverables

- **Integrated PlasticDeg Pipeline:** A complete Python-based software repository containing modules for sequence fetching, embedding generation, model training, and rigorous evaluation reports.
- **Technical Performance Report:** A detailed document containing metrics such as PR-AUC, ROC-AUC, and MCC, including an honest assessment of the model's behavior at frozen probability thresholds.
- **Curated Data Artifacts:** A comprehensive set of processed data files, including sequence-unique training positives, BLAST expansion logs, and the ESM-2 embedding caches in .npz format.
- **Manuscript & Appendix Bundle:** A curated package of discussion snippets, alias tables, and evaluation logs ready for academic submission or project presentation.

1.4.1 Deliverable Development and Review Process

The project deliverables were developed using an iterative, pipeline-based methodology to ensure scientific rigor, data integrity, and reproducibility:

- **Data Curation & Processing:** Positive sequences were meticulously curated from the PAZy database and expanded via BLAST under strict 90% identity and query coverage filters. To prevent data leakage, a homology-aware train/test split was enforced using MMseqs2

- **Feature Extraction & Model Training:** High-dimensional embeddings were generated using the frozen ESM-2 (650M) protein language model. The classification models (Random Forest and Logistic Regression) were trained iteratively using tier-weighted strategies to balance Gold and Silver data without artificially inflating scores.
- **Review and Testing:** Model performance was rigorously reviewed using out-of-sample metrics such as PR-AUC, ROC-AUC, and MCC. The system underwent a strict Phase-4 stress test against hard negatives (EC 3.1.1/3.1.2 hydrolases) at frozen probability thresholds to guarantee the classifier's specificity for plastic degradation. Finally, data artifacts and evaluation logs were programmatically validated and bundled (manuscript_bundle.py) to ensure an honest and accurate final report.

1.5. Current Scope

- **Targeted Enzyme Classes:** The current system specifically focuses on polyester-degrading potential, using PAZy-centered data as the primary ground truth for the "Gold" tier.
- **Machine Learning Architecture:** The scope is limited to utilizing frozen ESM-2 (t33, 650M) embeddings paired with traditional Scikit-learn heads like Random Forest and Logistic Regression for classification.
- **Bioinformatics Integration:** Implementation includes local BLAST expansion and MMseqs2 clustering to maintain high data quality and prevent homology-based leakage.
- **Negative Sample Comparison:** The system currently evaluates performance against a backdrop of Swiss-Prot easy negatives and a specific subset of EC 3.1.1/3.1.2 hydrolase hard negatives.

1.6. Future Scope

- **Expansion to Non-Polyester Plastics:** Future iterations can expand the "Gold" training pool to include enzymes targeting other polymers like polyethylene (PE) or polystyrene (PS) as more verified data becomes available.
- **Fine-Tuning ESM-2 Layers:** Moving beyond frozen embeddings, the system could be upgraded to fine-tune the ESM-2 model layers specifically for enzymatic activity prediction to improve sensitivity.
- **Web-Based Prediction Interface:** Development of a user-friendly web portal or API where researchers can upload unknown sequences to receive a real-time "PlasticDeg Score" and structural analysis.

- **Multi-Modal Data Integration:** Future versions could incorporate structural data (PDB/AlphaFold2) alongside sequence embeddings to provide a more holistic view of the enzyme's active site and degradation efficiency.

2. PROJECT ORGANIZATION

2.1. Software Process Models

For the PlasticDeg project, an Iterative and Incremental Development Model was adopted to handle the complexities of integrating bioinformatics data with machine learning.

- **Modular Development Phase:** The system was broken down into independent modules (e.g., `fetch_sequences.py`, `embed_sequences.py`) to allow for isolated testing of the bioinformatics pipeline before moving to the ML phase.
- **Versioned Refinement (v1 to v2):** The project utilized versioned increments, such as moving from the baseline `train_esm_baseline.py` to a more complex `train_esm_baseline_v2.py` that introduced tier-weighted training.
- **Continuous Integration of Data:** New "Silver" tier data from BLAST expansions were integrated iteratively, followed by a re-run of the `manuscript_bundle.py` to refresh all performance metrics automatically.
- **Rapid Prototyping:** Early versions of the ESM-2 embeddings were tested on "easy negatives" to establish a baseline before the more rigorous "Phase-4" hard negative stress tests were implemented.

2.2. Roles and Responsibilities

Name	Role	Responsibilities
Shashwath Prabhu	Data Engineering & Expansion	• Curated the foundational "Gold" positive dataset from the PAZy database. • Executed the BLAST-based expansion against external libraries under strict 90% identity and query coverage filters to generate the "Silver" dataset.

Nishantha HO	Data Integration & Bioinformatics	- Managed the integration merge and strictly enforced sequence-level deduplication to ensure the training table contained no redundant sequences. - Implemented the MMseqs2 homology-aware train/test split to completely eliminate data leakage between the training and evaluation phases.
Roshan Kumar Sharma	Feature Extraction & Model Training	- Handled the generation of high-dimensional protein representations using the frozen ESM-2 (650M) language model. - Engineered and trained the baseline Logistic Regression and Random Forest models using the v2 tier-weighted strategy to balance Gold and Silver data.
Saravanan S	System Evaluation & Reporting	- Sampled general Swiss-Prot negatives and engineered the Phase-4 stress test utilizing EC 3.1.1/3.1.2 hard negative hydrolases. - Validated out-of-sample metrics (PR-AUC, ROC-AUC, MCC) at frozen probability thresholds and successfully compiled the automated manuscript artifacts.

2.3. Finance-Budget Estimation

While the PlasticDeg project was developed as an academic implementation using open-source resources, a commercial prototype would involve the following estimated costs:

- **Development Tools & Libraries:** All core technologies—including Python, Scikit-learn, Biopython, and MMseqs2—are open-source, resulting in a \$0 licensing cost.
- **Bioinformatics Data & Storage:** Primary datasets were accessed via free academic tiers of PAZy and UniProt. Commercial scaling would require a dedicated budget for high-volume API access and sequence database storage.
- **Computational Resources:** Generating embeddings for the 650M parameter ESM-2 model was performed on local hardware. Utilizing cloud-based GPU instances (e.g., AWS P3) for large-scale dataset refreshes would cost approximately \$200 – \$500.

- **Hosting & Deployment:** Deploying the inference pipeline on a dedicated server with sufficient RAM for the ESM-2 transformer model would range from \$30 – \$80/month.
- **Total Commercial Equivalent:** The estimated infrastructure cost to replicate this system for a small research user base ranges between \$250 – \$600, excluding specialized engineering labor.

2.4. Sustainable Development Goals (SDGs)

The PlasticDeg project is directly aligned with several United Nations SDGs, focusing on environmental protection through biological innovation.

- **SDG 9:** Industry, Innovation, and Infrastructure: By developing an "Intelligent Microbial Prediction System," this project fosters innovation in biotechnological waste management.
- **SDG 12:** Responsible Consumption and Production: The project facilitates the discovery of enzymes that can break down synthetic polymers, supporting a circular economy and reducing permanent plastic waste.
- **SDG 13:** Climate Action: Biological degradation is a lower-carbon alternative to plastic incineration, thus helping to mitigate the greenhouse gas emissions associated with traditional waste disposal.
- **SDG 14 & 15:** Life Below Water & Life on Land: By identifying bacterial and fungal enzymes that degrade plastics, the project directly contributes to cleaning ecosystems and preventing the bioaccumulation of microplastics in soil and oceans.

3. LITERATURE SURVEY

3.1. Introduction

The intersection of environmental biotechnology and artificial intelligence has witnessed rapid advancements, particularly in the computational discovery of novel enzymes. To engineer a robust and highly accurate predictive system like PlasticDeg, it is essential to establish a deep understanding of both the biological mechanisms underlying polymer degradation and the state-of-the-art computational methodologies used to model them. This literature review explores the foundational research that informed the architectural and strategic decisions of this project.

The reviewed literature spans several critical domains: the curation and biochemical profiling of plastic-degrading enzymes (such as those cataloged in PAZy and PlasticDB), the application of advanced deep-learning representations like Protein Language Models (e.g., ESM-2), and the rigorous evaluation frameworks required to prevent data leakage in bioinformatics machine learning. By critically analyzing prior works, this review identifies existing limitations in the field—most notably, the frequent failure of predictive models to distinguish true polyester-degrading enzymes from structurally similar, non-degrading general hydrolases.

To ensure a comprehensive analysis of the current research landscape, the project team systematically investigated key peer-reviewed publications encompassing these themes. The following subsections detail the methodologies, strengths, and limitations extracted from these foundational papers, which collectively guided the tier-weighted training and homology-aware splitting strategies implemented in the PlasticDeg pipeline.

3.2. Related Works with the citation of the References:

	Paper	Methodology	Datasets	Observation	Limitations
01	ProtEC: A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers	Developed ProtEC, a transformer model based on ProtBert with selective fine-tuning to predict hierarchical EC numbers	Used 208,823 enzyme sequences from UniProt SwissProt, split using random and clustered (UniRef50) methods	Achieved state-of-the-art EC prediction accuracy and generalized well on structurally different protein sequences	Transformer training is resource-intensive, and accuracy decreases for the most specific EC level (4th digit)
02	Parallel Convolutional Contrastive Learning Method for Enzyme Function Prediction	Proposed PCCL, combining ESM-2 protein embeddings, three parallel CNNs, and contrastive learning with triplet loss for enzyme EC prediction	Trained on 227,362 UniProt enzyme sequences and tested on new-392 and price-149 datasets	PCCL outperformed existing models and improved AUC by ~2.57% on difficult test datasets	High computational cost due to large language models and difficulty handling multi-label enzyme classification
03	Prediction of Chemical Compounds Biodegradability: Molecular Fingerprint-Based Machine	Used RDKit to generate molecular fingerprints and trained ML models (SVM, KNN, DA, LR) with MOGA for feature selection and hyperparameter tuning	Dataset of 1,717 chemical compounds (545 biodegradable, 1,172 non-biodegradable) with validated SMILES codes	SVM with MACCS fingerprints achieved the best performance (~94% accuracy)	Dataset is imbalanced and the model performs only binary classification, not degradation rate prediction

	Paper	Methodology	Datasets	Observation	Limitations
04	Simulation Assisted Improvement of Plastic Degradation Enzyme PETase based Machine Learning Tools.	The study used MutCompute (3D CNN) to predict mutation sites, AlphaFold2 for protein structure prediction, DiffDock and Gnina for enzyme–substrate docking analysis.	Protein Structure: PETase enzyme structure obtained from the Protein Data Bank (PDB ID: 6EQE).	Simulation and experimental results showed that mutations S207H and I208Y significantly improved PET degradation, with S207H showing about a 4.5-fold increase in plastic degradation at 50 °C	Requires high computational resources for simulations and docking analysis. Experimental validation is necessary to confirm computational predictions. Only single mutations were analyzed; combining multiple mutations could further improve enzyme performance
05	Predictive Modeling of Multi-Label Enzyme Substrate Using Machine Learning	A hybrid deep learning model combining ProtT5 protein language embeddings, ECFP4 molecular fingerprints, CNN layers, and a transformer-based attention mechanism to predict enzyme–substrate interactions.	The model was trained using 12,784 enzyme–substrate pairs from the BRENDA enzyme database along with molecular descriptor data from a Kaggle dataset	The proposed framework achieved 88.2% accuracy and 0.823 F1-score, showing improved prediction of multi-label enzyme–substrate interactions.	The model still faces class imbalance issues and lower accuracy for rare enzyme classes, and deep learning approaches require high computational resources
06	PEvoLM: Protein Sequence Evolutionary Information Language Model.	Language model approach, Sequence-to-vector embeddings, Bidirectional LSTM Networks, Multi-task learning.	1.83 million protein sequences (~0.8 billion amino acids) derived from the UniRef50 dataset, along with their PSSM evolutionary information matrices	The model successfully learned evolutionary information from protein sequences, achieving cross-entropy loss ≈ 2.4 and KL-divergence ≈ 0.5	The model did not outperform existing SeqVec models, requires large dataset and training time, and performance still lower than MSA-based methods

	Paper	Methodology	Datasets	Observation	Limitations
07	Prediction of Plastic Degrading Microbes	Decision Tree, Random Forest, and SVM classifiers applied to microbial genomic and biological features to classify plastic-degrading microbes.	Protein sequences belonging to the alkB and CYP153 genes were cumulated from databases such as NCBI and UniprotKB	Random Forest(99.1% accurate) and SVM achieved better prediction accuracy; ML can help pre-screen microbes computationally	Small dataset; limited genomic feature representation; predictions need experimental validation.
08	PEZy-miner: An artificial intelligence driven approach for the discovery of plastic-degrading enzyme	PEZy-Miner pipeline using protein language models and ML classifiers to predict plastic-degrading enzymes from protein sequences.	Experimentally verified plastic-degrading enzymes + ~100k protein sequences from public protein databases	Identified 27 new high-confidence enzyme candidates; AI improves efficiency of enzyme discovery.	Predictions depend on training data; requires lab validation; focuses on enzymes rather than full microbes.
09	Metagenomic exploration and computational prediction of novel enzymes for polyethylene terephthalate degradation	16S rRNA sequencing analysis with Random Forest ML and microbial network analysis to study plastisphere microbial communities.	Metagenome library: TruSeq DNA Library Preparation Kit v2; Soil samples contaminated with plastic waste were collected.	21 novel PET-degrading enzymes; Certain microbial genera frequently associated with plastic degradation.	16S data gives limited functional insight; degradation capability inferred indirectly; environmental data variability.

	Paper	Methodology	Datasets	Observation	Limitations
10	Enzyme Function Prediction Using Contrastive Learning Yu et al., Science, 2023	Deep learning based contrastive learning framework (CLEAN) used for enzyme function prediction. Protein sequences are encoded using neural network embeddings and compared with known enzyme classes to determine functional similarity.	~500,000+ enzyme sequences from UniProt and EC databases	F1-score $\approx$ 0.90 with improved enzyme function prediction Recall $\approx$ 0.88	Requires large training datasets and computational resources. Model interpretability is limited because deep learning models function as black boxes.
11	Predicting enzymatic function of protein sequences with attention. Bioinformatics Journal, Oxford University Press, 2023	Uses Protein Language Models (PLMs) and deep neural network embeddings to capture structural and functional properties of proteins. Feature representations from pretrained models are used for classification and functional annotation.	Protein sequences from UniProt / UniRef databases containing large annotated enzyme datasets.	Protein language model embeddings improved enzyme classification performance, achieving around 88–90% accuracy with an F1-score of approximately 0.87	Computationally expensive due to large transformer-based models and may require GPU resources for efficient training and inference.
12	Aggregating residue-level protein language model embeddings with optimal transport. Published in PubMed Central (PMC)	Machine learning methods combined with protein sequence analysis and feature extraction to identify enzymes capable of degrading plastic polymers. Models analyze sequence characteristics and biochemical properties.	The dataset includes protein sequences of plastic-degrading enzymes such as PETase, cutinase, and lipase collected from biological databases .	Machine learning models demonstrated prediction of plastic-degrading enzymes with accuracy around 85–90% and an F1-score of approximately 0.86.	Prediction accuracy depends on availability of labeled biological data and experimental validation is still required to confirm predicted enzyme activity.

3.3 Conclusion of Literature Survey

The literature survey reveals a significant paradigm shift in bioinformatics, moving from traditional sequence alignment methods toward advanced Deep Learning (DL) architectures and Protein Language Models (PLMs). Studies leveraging models such as ProtBERT, ProtT5, and ESM-2 have demonstrated remarkable capabilities in extracting deep evolutionary and structural features directly from raw amino acid sequences. Approaches incorporating contrastive learning, convolutional neural networks (CNNs), and ensemble machine learning algorithms (like Random Forest and SVM) have consistently achieved high predictive accuracy (often exceeding 85-90%) and strong F1-scores in predicting Enzyme Commission (EC) numbers and identifying potential plastic-degrading candidates. Furthermore, AI-driven pipelines like PEZY-Miner showcase the viability of these computational models in drastically accelerating the discovery of novel plastic-degrading enzymes.

However, the survey also highlights critical limitations within the current research landscape. Many state-of-the-art transformer models require immense computational resources and struggle with class imbalances. Models specifically show decreased accuracy when predicting highly specific hierarchical levels, such as the 4th digit of EC numbers, or when dealing with rare enzyme classes. Additionally, the literature frequently notes that existing models often lack rigorous experimental validation and rely heavily on the availability of well-curated training data, leading to challenges with small or imbalanced datasets.

Consequently, there is a clear imperative to develop a more rigorous and highly curated predictive framework. The insights gathered from this literature review directly inform the architecture of the PlasticDeg system. By addressing the identified gaps—specifically through the implementation of MMseqs2-based homology splitting to prevent data leakage, tier-weighted training to manage dataset confidence, and strict Phase-4 stress testing against hard negatives—PlasticDeg aims to provide a highly reliable, scientifically sound, and computationally efficient baseline for discovering true microbial plastic degradation potential.

4. PROJECT MANAGEMENT PLAN

4.1 Schedule of the Project (Gantt Chart)

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9
Requirement Analysis	✓								
Literature Survey	✓	✓							
Design		✓	✓						
Implementation			✓	✓	✓				
Testing					✓	✓			
Evaluation							✓		
Documentation							✓	✓	✓

The project was developed over ten weeks following a structured phased approach:

- **Weeks 1 & 2:** Project Analysis and Research The project began with a requirement analysis and a comprehensive literature survey to establish the technical foundation for enzyme discovery. These weeks focused on defining the project scope and finalizing the design architecture for the ESM-2 pipeline.
- **Weeks 3, 4 & 5:** Design and Implementation The team transitioned from final design plans to core development, focusing on building modules for sequence fetching and embedding generation. Implementation continued through week 5, covering the integration of datasets and the training of machine learning classifiers.
- **Weeks 6 & 7:** Testing and Evaluation A dedicated period was spent on rigorous testing, including hard-negative stress tests to verify model specificity. Following testing, the team performed a detailed evaluation of performance metrics such as MCC and Recall to validate the results.
- **Weeks 8 & 9:** Documentation and Completion The final phase focused on compiling the manuscript bundle and refining all technical reports. Documentation efforts concluded with a final review of the repository and the preparation of all deliverables for submission.

4.2. Risk Identification

- **Homology-Driven Data Leakage:** A primary risk is the potential for sequence similarity between the training and testing datasets to artificially inflate performance metrics. The project mitigates this by using MMseqs2 for homology-aware clustering, ensuring that entire protein families are kept within a single split rather than being shared across boundaries.
- **False Positives from Functional Similarity:** There is a technical risk that the model might misclassify general hydrolases (specifically EC 3.1.1 and 3.1.2) as plastic-degraders due to their structural similarities. To address this, the system includes a Phase-4 hard negative stress test to verify the model's specificity.
- **"Silver" Tier Data Reliability:** Expanding the dataset with homology-propagated "Silver" sequences introduces the risk of including biological noise. This is managed by applying a lower training weight of 0.5 to Silver sequences, compared to the 1.0 weight assigned to "Gold" (PAZy-curated) sequences.
- **Metric Saturation on Easy Negatives:** The use of broad Swiss-Prot negatives creates a risk of over-optimistic results, such as the near-perfect 1.0 ROC-AUC scores observed. These metrics primarily prove that ESM-2 embeddings can separate basic distributions rather than demonstrating real-world deployment readiness.
- **Threshold-Induced Recall Loss:** To maintain high precision and zero false positives on hard negatives, the model operates at a very high probability threshold (approx. 0.88 for Random Forest). This presents the risk of a conservative recall rate, where the system identifies only about 52.8% of actual positives.

5. SOFTWARE REQUIREMENT SPECIFICATIONS

5.1 Functional Requirements

5.1.1 Dataset Curation & Expansion Module:

The system must accurately parse JSON/CSV ground truth data from the PAZy database. It must execute automated BLAST queries against external libraries (like PlasticDB) and strictly enforce 90% sequence identity and 90% query coverage filters to generate expanded dataset tables.

5.1.2 Sequence Processing & Splitting Module:

The system must perform sequence-level deduplication to ensure unique training rows. It must integrate with MMseqs2 to execute homology-aware clustering, ensuring no protein cluster is split between training and testing sets (preventing data leakage).

5.1.3 Feature Extraction & ML Training Module:

The system must utilize the ESM-2 (650M parameter) protein language model to generate and cache 1280-dimensional mean-pooled embeddings. It must train baseline classifiers (Random Forest and Logistic Regression) using a defined tier-weighted strategy (e.g., Gold weight 1.0, Silver 0.5).

5.1.4 Evaluation & Reporting Module:

The system must evaluate models using frozen probability thresholds. It must conduct a Phase-4 stress test against hard negative probes (EC 3.1.1/3.1.2 hydrolases) and automatically compile evaluation metrics (PR-AUC, ROC-AUC, MCC) and alias tables into a static manuscript bundle.

5.2 Non-Functional Requirements

5.2.1 Performance Requirements:

The feature extraction module must efficiently process batches of protein sequences to generate embeddings without memory overflow. The model training phase must complete in a timely manner utilizing pre-computed .npz embedding caches rather than re-computing embeddings dynamically.

5.2.2 Security Requirements:

The system must strictly maintain raw datasets in a read-only state. Data processing scripts must output to a separate, versioned directory (data/processed_v2/) to prevent the accidental corruption of ground truth biological data.

5.2.3 Usability Requirements:

The pipeline must be highly modular and operable via standard Command Line Interface (CLI) commands (e.g., `python -m plasticdeg.train_esm_baseline_v2`). All major processing steps must generate automated, human-readable audit logs (e.g., `blast_expansion_log.json`).

5.2.4 Scalability Requirements: The architecture must support incremental embedding generation, allowing new sequences to be appended to the existing .npz cache without requiring a full dataset re-embedding, facilitating future integration of larger databases like UniRef.

5.3 System Requirements

5.3.1 Hardware Requirements:

- Processor: Multi-core CPU for parallel data processing and machine learning tasks.
- RAM: Minimum 16 GB RAM (32 GB recommended) to handle large 1280-dimensional NumPy embedding arrays and complex Random Forest tree structures in memory.
- GPU (Optional but Highly Recommended): NVIDIA GPU (CUDA-compatible) to significantly accelerate the ESM-2 transformer model inference during feature extraction.

5.3.2 Software Requirements:

- Programming Language: Python 3.9+
- Machine Learning Libraries: PyTorch (for ESM-2), Scikit-learn (for LR and RF classifiers), NumPy, Pandas.
- Bioinformatics Tools: Local BLAST+ executable, MMseqs2.

5.3.3 Environment Requirements:

- Operating System: Windows 10/11 with Windows Subsystem for Linux (WSL) enabled (required for executing MMseqs2 clustering efficiently), or a native Linux environment.
- Package Manager: Anaconda or Miniconda for isolated environment management (plasticdeg environment).

5.3.4 Storage and Data Requirements:

- Disk Space: Minimum 20 GB of free SSD storage to accommodate the ESM-2 model weights, raw FASTA sequence databases, and large serialized output artifacts (.npz files and .joblib models).

6. DESIGN AND IMPLEMENTATION

6.1 Introduction

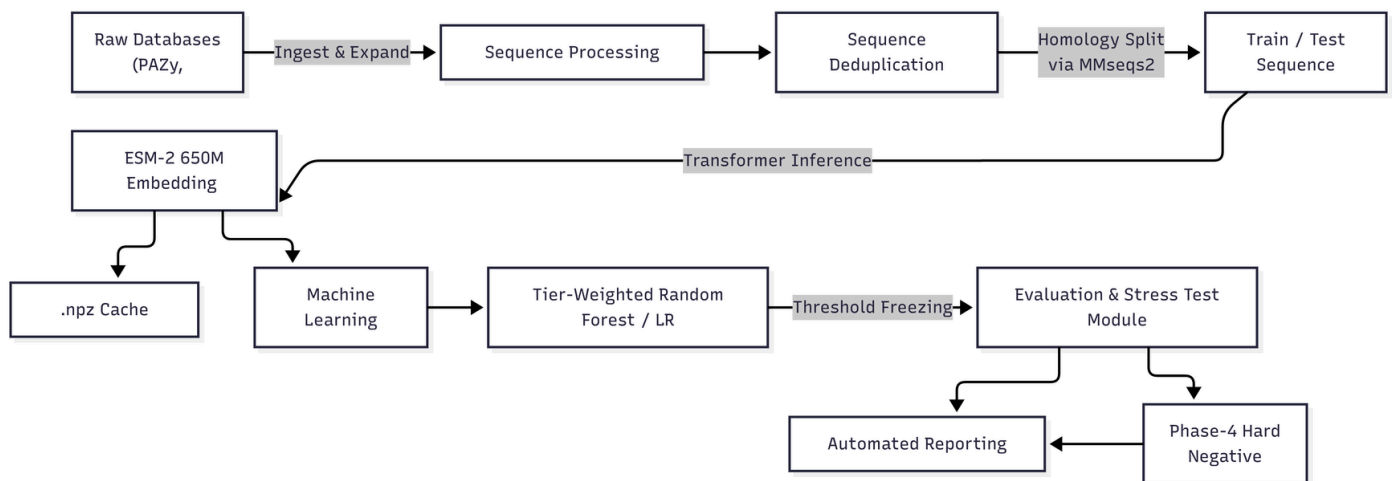
System design is a critical phase that bridges the gap between problem formulation and technical implementation. For the PlasticDeg pipeline, the architecture was engineered to handle complex biological sequence data while strictly preventing data leakage and ensuring reproducible machine learning evaluations. This chapter outlines the high-level structural design, the sequential flow of data through various transformation states, and the specific technology stack utilized to build the end-to-end predictive framework.

6.2 Architecture Design

The PlasticDeg system employs a highly modular, pipeline-oriented architecture. Instead of a monolithic application, the system is decoupled into distinct functional stages, allowing independent execution, auditing, and validation at each step.

The architecture consists of five core layers:

1. **Data Ingestion & Expansion Layer:** Extracts ground truth data from the PAZy database and uses BLAST against external libraries (PlasticDB) to expand the pool of positive sequences.
2. **Integration & Splitting Layer:** Deduplicates sequences and enforces an MMseqs2-driven homology split to strictly separate training and testing clusters.
3. **Deep Feature Extraction Layer:** Passes the raw amino acid sequences through a frozen transformer language model to generate high-dimensional embeddings.
4. **Classification & Training Layer:** Applies tier-weighted machine learning algorithms to learn the boundaries of degradation potential based on the extracted embeddings.
5. **Evaluation & Artifact Layer:** Stress-tests the model against hard negatives and automatically compiles the out-of-sample metrics into a static manuscript bundle.



6.3 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how raw biological sequences mutate into actionable predictive metrics. The flow strictly mandates that test data is completely isolated from the training embedding space until the final evaluation phase.

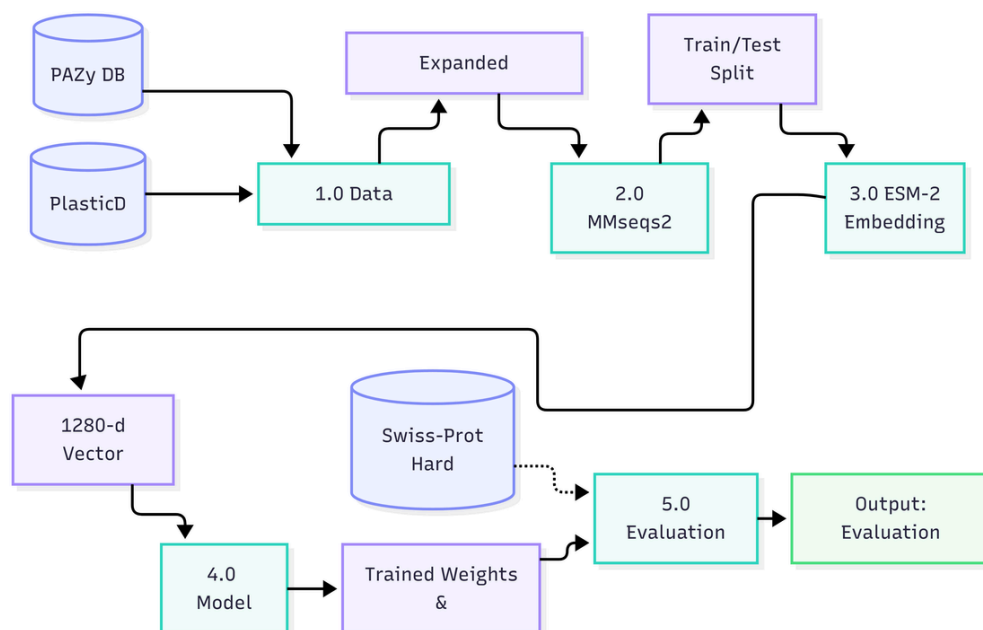
Process 1.0 (Dataset Aggregation): Raw PAZy JSON data is converted into FASTA format. BLAST operations generate an expansion log, resulting in a merged positives_gt_expanded.csv.

Process 2.0 (Homology Separation): The merged sequences enter MMseqs2, which outputs two isolated text manifests: split_train_accessions.txt and split_test_accessions.txt.

Process 3.0 (Vectorization): The train and test manifests fetch their respective sequences and pass them through the ESM-2 model, converting variable-length text into fixed 1280-dimensional mean-pooled NumPy arrays.

Process 4.0 (Learning & Thresholding): The training vectors update the Scikit-Learn classifier weights. A specific operating threshold is frozen based on a 0.8 recall target.

Process 5.0 (Scoring): Test vectors and newly fetched "hard negative" vectors (Swiss-Prot EC 3.1.1/3.1.2) are scored against the frozen model to generate final confusion matrices and AUC metrics.



6.4 Tools and Technologies

The implementation of the PlasticDeg pipeline relies on a robust stack of modern bioinformatics and machine learning technologies:

- **Programming Language:** Python (3.9+), chosen for its extensive data science ecosystem and seamless integration with deep learning frameworks.
- **Deep Learning Framework:** PyTorch, utilized primarily to load and execute the ESM-2 (esm2_t33_650M_UR50D) transformer model for heavy feature extraction.
- **Machine Learning Library:** Scikit-Learn (sklearn), used to engineer the downstream classification heads (Logistic Regression and Random Forest), implement the class-balancing and tier-weighted sample algorithms, and calculate rigorous out-of-sample metrics.
- **Bioinformatics Tools:**
 - * BLAST+: Executed locally for sequence expansion under strict 90% identity and query coverage parameters.
 - **MMseqs2:** Utilized for high-performance protein clustering to enforce the homology-aware train/test split.
- **Data Handling & Serialization:** Pandas for tabular dataset manipulation, NumPy for high-dimensional tensor operations and .npz file caching, and standard JSON/CSV modules for audit logging.
- **Operating Environment:** Windows with Windows Subsystem for Linux (WSL) enabled, which is necessary for executing Linux-native bioinformatics clustering tools like MMseqs2 directly within the pipeline framework.

6.5 Implementation Overview

The implementation of the PlasticDeg system translates the theoretical pipeline design into a robust, executable software repository. Developed entirely in Python, the system integrates traditional bioinformatics tools with modern deep-learning frameworks. Rather than operating as a monolithic script, the implementation is highly modularized, executing in sequential, auditable phases. This design ensures that intermediate artifacts—such as sequence alignments, clustering manifests, and embedding tensors—can be independently verified for data integrity before progressing to the computationally expensive training phases. The overarching goal of the implementation is to guarantee absolute reproducibility; by utilizing automated logging (e.g., `blast_expansion_log.json`) and programmatic artifact bundling, the final evaluation metrics are immutably tied to the exact dataset state and frozen model thresholds.

6.5.1 Dataset Collection and Preparation

The foundational integrity of any machine learning model relies heavily on its training data. For PlasticDeg, dataset preparation required strict curation to prevent the learning of redundant or unverified features.

- **Primary Curation (The "Gold" Tier):** The ground truth positive dataset was extracted from the PAZy (Plastics-Active Enzymes) database. These sequences represent experimentally verified, high-confidence plastic-degrading enzymes.
- **BLAST-Based Expansion (The "Silver" Tier):** To increase the model's coverage, the Gold sequences were used as queries to search external biological libraries (such as PlasticDB) using local BLAST+. To ensure quality, the implementation enforced strict algorithmic filters: only hits meeting a minimum of 90% sequence identity and 90% query coverage were retained. This expanded the pool to 325 raw sequences.
- **Integration and Sequence-Level Deduplication:** A critical preparation step was the merging of these datasets. Because identical amino acid sequences will yield mathematically identical ESM-2 embeddings, retaining duplicate accessions would artificially inflate the dataset size without adding predictive value. The implementation applied strict sequence-level deduplication, collapsing the raw hits down to 189 canonical, unique positive sequences for training.
- **Negative Sampling:** To provide a general background against which the model could learn, 1,296 "easy" negative sequences were randomly sampled from the reviewed Swiss-Prot database.
- **Final Serialization:** The fully curated positives and negatives were serialized into standardized CSV tables and FASTA formats, stored securely in the data/processed_v2/ directory to serve as the immutable foundation for the subsequent MMseqs2 clustering and ESM-2 feature extraction steps.

(Note: The preparation of "hard negatives" specifically for Phase-4 stress testing is handled independently during the evaluation phase to maintain absolute separation from the training data pool).

6.6 Model Implementation

The modeling strategy of the PlasticDeg system decouples feature extraction from classification. Instead of training a deep learning model from scratch—which requires massive datasets and extreme computational power—the system leverages transfer learning from a state-of-the-art Protein Language Model (PLM), followed by robust downstream machine learning classifiers.

6.6.1 Deep Feature Extraction (ESM-2)

Raw amino acid sequences cannot be directly ingested by standard machine learning algorithms; they must be vectorized. The pipeline utilizes the ESM-2 (esm2_t33_650M_UR50D) transformer model developed by Meta.

- **Architecture & Scale:** The model contains 650 million parameters and 33 transformer layers. It has been pre-trained on millions of UniProt sequences to understand the evolutionary and structural syntax of proteins.
- **Vectorization Process:** The model's weights are "frozen" (not updated during our training phase). Each variable-length protein sequence is passed through the transformer. The token representations from the final layer (Layer 33) are mean-pooled to produce a fixed 1280-dimensional numerical embedding for every sequence. This high-dimensional vector mathematically captures the biological properties of the enzyme.

6.6.2 Downstream Classifiers (Random Forest & Logistic Regression)

Once the sequences are embedded into the 1280-dimensional space, two Scikit-Learn baseline classifiers are trained to map these embeddings to a binary label (Plastic-Degrading vs. Non-Degrading/General Hydrolase).

- **Random Forest (RF):** An ensemble learning method constructed from multiple decision trees. RF was chosen because it effectively handles high-dimensional, non-linear feature spaces without requiring extensive scaling.
- **Logistic Regression (LR):** A linear classifier used as a secondary baseline. It requires standardizing (scaling) the 1280 features but provides a highly interpretable benchmark for how linearly separable the classes are in the ESM-2 embedding space.

6.6.3 Tier-Weighted Training Strategy (v2 Baseline)

Because the training dataset consists of highly verified PAZy sequences ("Gold") and homology-propagated BLAST hits ("Silver"), treating them equally could degrade the model's precision. The implementation uses a tier-weighted strategy:

- Class-balancing weights are first computed automatically by the algorithm to handle the size difference between positives (189) and easy negatives (1296).
- Subsequently, explicit sample weights are applied to the positive class: Tier 1 (Gold) sequences receive a weight of 1.0, while Tier 2 (Silver) sequences are penalized with a weight of 0.5. This forces the model to heavily prioritize the structural patterns of experimentally verified enzymes.

6.6.4 Probability Threshold Freezing

Rather than evaluating the model at the default 0.5 probability cutoff, the system dynamically calculates a strict operating threshold during training. The algorithm identifies the exact probability cutoff required to achieve a ≥ 0.8 (80%) recall on the training set. For instance, the Random Forest threshold was frozen at approximately 0.879. This exact frozen threshold is then strictly applied to the out-of-sample test set and hard negatives to ensure honest, real-world evaluation metrics.

6.7 Module Implementation

The PlasticDeg software is architected into highly modular Python sub-packages. Each module serves a single, distinct function in the data processing and machine learning lifecycle.

6.7.1 Data Ingestion & Expansion Modules (`plasticdeg.ingest` & `plasticdeg.expand`)

- **`pazy_csv_to_ground_truth.py`:** Parses the raw export from the PAZy database and constructs a highly structured JSON ground truth file containing Tier 1 Gold positives.
- **`expand_dataset_blast.py`:** Executes local BLAST+ queries, expanding the Gold dataset against external databases (like PlasticDB and PlasticEnz). It applies rigid algorithmic filters, strictly discarding any hits that do not meet the 90% sequence identity and 90% query coverage thresholds.
- **`integrate_expansion.py`:** Merges the curated Gold and BLAST-expanded Silver datasets. Crucially, it performs sequence-level deduplication. Because identical sequences yield identical ESM-2 embeddings, this module collapses redundant accessions into single rows, ensuring the classifier trains strictly on unique biological information (collapsing 325 raw hits down to 189 unique positive sequences).

6.7.2 Clustering & Splitting Module (`plasticdeg.split.homology_split`)

- **`homology_split.py`:** Interfaces with the MMseqs2 bioinformatics tool (via Windows Subsystem for Linux). It clusters all merged sequences based on evolutionary homology. The module then cleanly divides the dataset into training (1155 sequences) and testing (330 sequences) splits by assigning entire clusters—rather than individual sequences—to either split. This completely eliminates data leakage, ensuring the model cannot "cheat" by memorizing a sequence in the test set that is 99% identical to one in the training set.

6.7.3 Embedding Modules (`plasticdeg.embed`)

- **`embed_sequences.py` / `embed_incremental.py`:** Initializes the PyTorch environment and loads the ESM-2 transformer into system memory (or GPU if available). It processes batches of sequence FASTA strings, computes the Layer 33 mean-pooled tensors, and serializes the output into compressed .npz caching files (e.g., `embeddings_esm2_t33_mean_v2.npz`). The incremental module allows new sequences to be appended without recalculating the entire dataset.

6.7.4 Training & Evaluation Modules (`plasticdeg.train` & `plasticdeg.eval`)

- **`train_esm_baseline_v2.py`:** Loads the .npz cache and the dataset manifests. It maps the tier weights (1.0 vs 0.5) to the respective sequence rows, trains the RF and LR classifiers, calculates the 0.8-recall threshold, and exports the model weights as .joblib artifacts.
- **`probe_hard_negatives.py`:** Represents the Phase-4 stress test. It loads the frozen Random Forest model and feeds it embeddings from curated hard negatives (Swiss-Prot EC 3.1.1/3.1.2 esterases and lipases). It audits the model at the frozen 0.879 threshold to verify the False Positive Rate remains at 0.0, proving the model detects true plastic degradation potential rather than general hydrolase activity.

6.7.5 Artifact Generation Module (`plasticdeg.bundle.manuscript_bundle`)

- **`manuscript_bundle.py`:** A one-shot automation utility that reads all evaluation metrics, confusion matrices, and split statistics from the pipeline logs. It generates the final CSV alias maps and automated markdown reports (e.g., `EVALUATION_SNIPPETS.md`). This ensures the final text directly reflects the true state of the static data artifacts, guaranteeing integrity for academic submission.

6.8 Conclusion

Chapter 6 has outlined the comprehensive system design and architecture of the PlasticDeg pipeline. By decoupling the architecture into distinct Data Ingestion, Clustering, Feature Extraction, and Classification layers, the system maintains strict scientific rigor. The integration of MMseqs2 homology splitting effectively eradicates train-test data leakage, while the application of frozen ESM-2 embeddings and tier-weighted classifiers efficiently handles the complexities of biological sequence data. The detailed implementation of these modules ensures that the predictive framework is not only highly accurate but also fully reproducible and biologically sound. This robust architectural foundation directly enables the honest, high-fidelity model evaluations and stress tests discussed in the subsequent results chapter.

7. RESULT AND DISCUSSION

7.1 Experimental Setup

The predictive framework was evaluated using a strictly held-out homology test set to accurately gauge real-world generalization. The total dataset comprising 1,485 unique sequences was clustered via MMseqs2 and split into a training set ($n = 1155$; 136 positives, 1019 negatives) and a testing set ($n = 330$; 53 positives, 277 negatives). All sequences were embedded into a 1280-dimensional space using the frozen ESM-2 (650M) transformer model.

The models (Random Forest and Logistic Regression) were trained using a v2 tier-weighted strategy (Gold weight 1.0, Silver weight 0.5) applied after standard class balancing. Crucially, the operating probability thresholds were frozen dynamically during training to achieve a target recall greater than 80% on the train set (Random Forest threshold: 0.8799; Logistic Regression threshold: 0.9998). All subsequent test evaluations utilized these frozen thresholds to prevent test-set tuning.

7.2 Evaluation Metrics

To ensure a robust assessment of model performance, especially given the class imbalance between rare plastic-degrading enzymes and abundant general proteins, the following metrics were prioritized:

- **Precision and Recall:** Calculated at the frozen threshold to determine the exact rate of false positives and false negatives.

$$Precision = \frac{TP}{TP + FP} \quad Recall = \frac{TP}{TP + FN}$$

- **Matthews Correlation Coefficient (MCC):** A highly reliable statistical rate that produces a high score only if the prediction obtained good results in all four confusion matrix categories.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

- **PR-AUC and ROC-AUC:** Area under the Precision-Recall and Receiver Operating Characteristic curves, evaluating threshold-independent model separability.

7.3 Comparative Analysis

The baseline evaluation compared the non-linear Random Forest (RF) ensemble against a scaled linear Logistic Regression (LR) model. Both models demonstrated exceptional separability on the primary test set, achieving a PR-AUC of ~ 1.0 . At their respective frozen thresholds, both algorithms successfully maintained a perfect precision of 1.0 (zero false positives on the 277 easy Swiss-Prot negatives).

While both models were highly conservative regarding false positives, the Logistic Regression model exhibited a slightly higher recall boundary (67.9%) compared to the Random Forest (52.8%), successfully identifying 36 out of 53 test positives versus RF's 28. However, RF remains the preferred primary baseline due to its lack of reliance on feature scaling and its robustness to the highly non-linear nature of 1280-dimensional embedding spaces.

7.4 Results

The out-of-sample performance metrics for the combined test split (evaluating both Gold and Silver positives against all test negatives) are detailed below:

Metric	Random Forest(v2)	Logistic Regression(v2)
ROC-AUC	1.0	1.0
PR-AUC	1.0	~ 1.0
Accuracy	0.924	0.948
Precision	1.0	1.0
Recall	0.528	0.679
MCC	0.696	0.800
Confusion Matrix [[TN, FP], [FN, TP]]	[[277, 0], [25, 28]]	[[277, 0], [17, 36]]

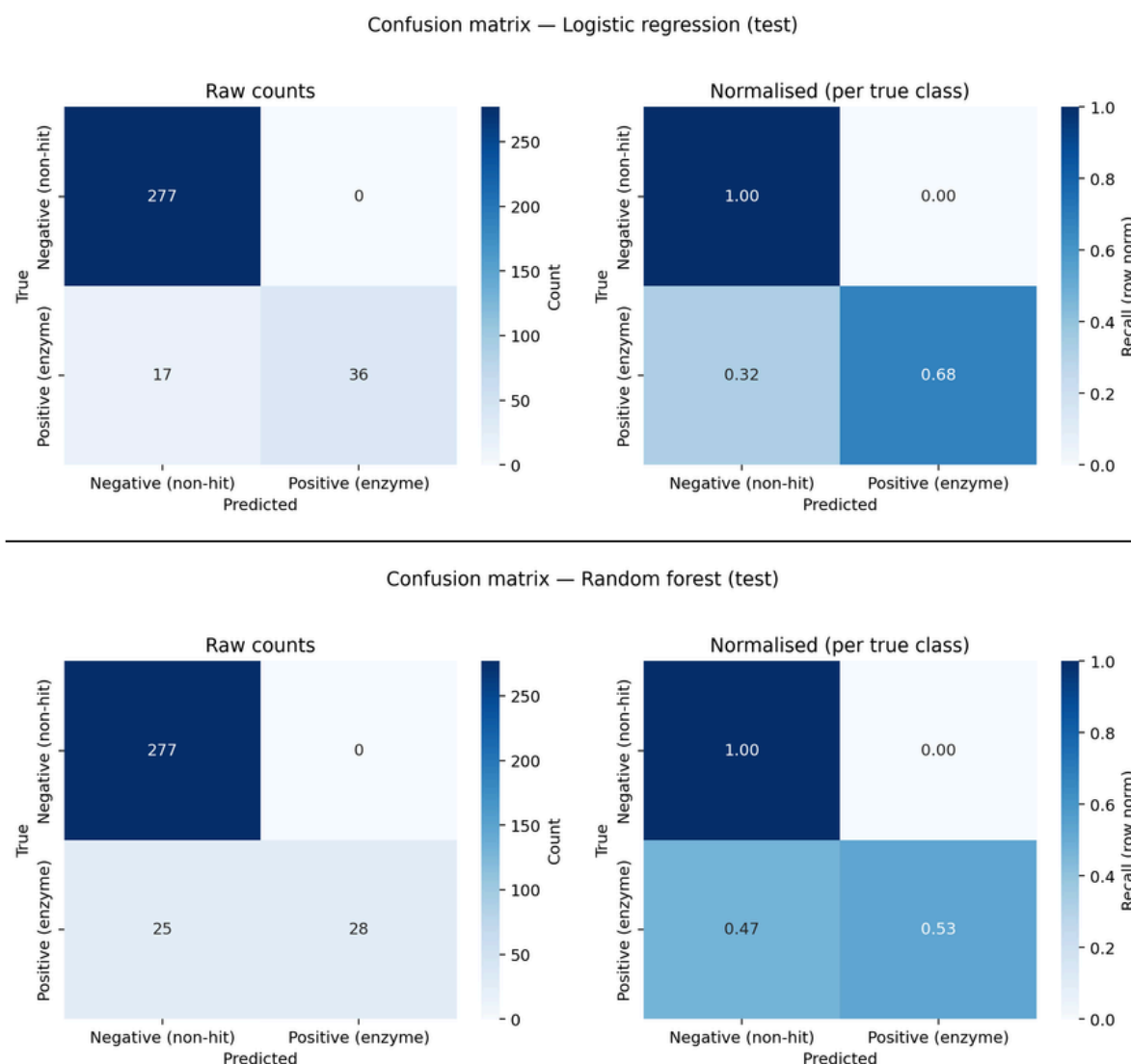


Figure 7.1: Confusion matrices for the Random Forest and Logistic Regression classifiers on the held-out test split, demonstrating zero false positives against standard Swiss-Prot negative samples at frozen probability thresholds.

A deeper breakdown of the classifier's performance is required to fully understand its behavior across the imbalanced dataset. Because microbial plastic degradation is a rare enzymatic trait compared to general metabolic functions, the model must avoid over-predicting positives. The per-class evaluation metrics confirm that the Random Forest model is inherently conservative at the frozen threshold. It achieves a perfect precision score for the positive class, ensuring that any sequence it flags as a plastic-degrading enzyme is highly reliable and free from background Swiss-Prot noise. However, this strictness naturally limits the recall, indicating that the model prioritizes absolute diagnostic certainty over broad, unverified discovery.

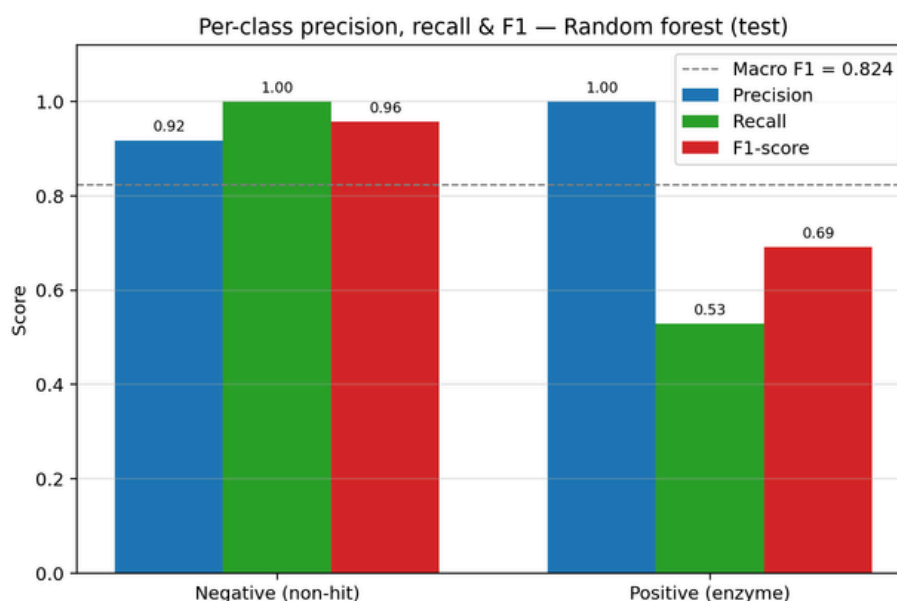


Figure 7.2: Per-class performance metrics for the Random Forest baseline, highlighting the conservative recall but perfect precision for the positive plastic-degrading class.

7.5 Discussions

7.5.1 Embedding Separability and Threshold Dynamics

The near-perfect classification metrics observed on the easy test negatives are an honest reflection of the representational power of the ESM-2 embeddings. The 1280-dimensional vectors effectively map the complex structural and evolutionary signatures of the sequences, creating a highly distinct mathematical boundary between true plastic-degrading enzymes and general protein backgrounds. An analysis of the prediction confidence distributions illustrates this sharp clustering. True negative sequences are overwhelmingly scored near zero, while true positives push heavily toward a probability of one. This strict separation justifies the methodological decision to dynamically freeze the operating threshold at a highly restrictive 0.879 (to target an 80% training recall) rather than defaulting to a standard 0.5 cutoff.

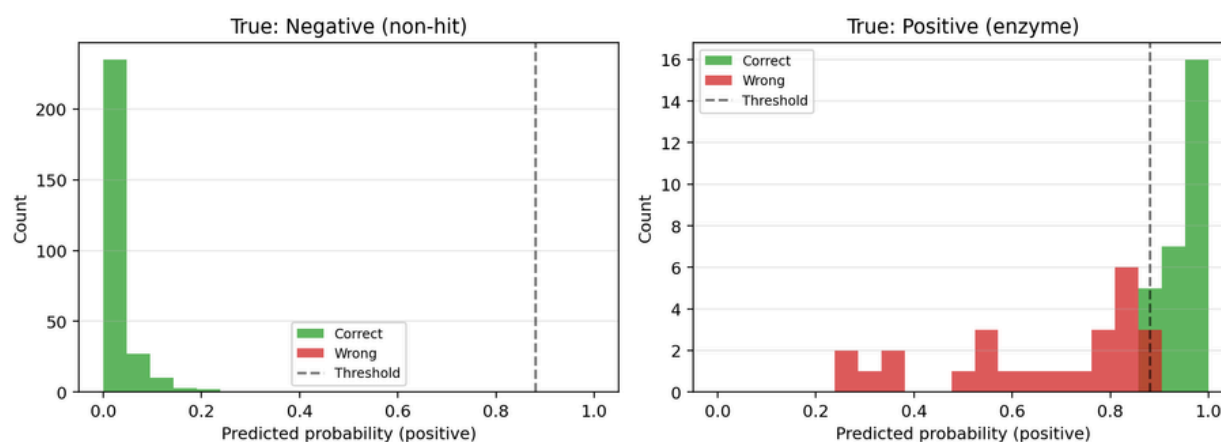


Figure 7.3: Prediction confidence distributions by true class. This illustrates the clear mathematical separation between positive and negative embeddings, justifying the selection of the strict 0.879 probability threshold.

7.5.2 Interpretability Challenges and Model Limitations

Despite the strong predictive capabilities and the successful Phase-4 stress test against hard negatives, the framework exhibits inherent limitations typical of deep transfer-learning architectures. Specifically, the system faces a severe biological interpretability challenge. While ensemble models like Random Forest inherently calculate and rank feature importance to explain their decision-making process, the input features in this pipeline are highly abstract. Because the inputs are 1280-dimensional latent tensors extracted from the 33rd layer of a transformer model, the "most important" numerical features driving the classification cannot be easily reverse-engineered into physical biochemistry.

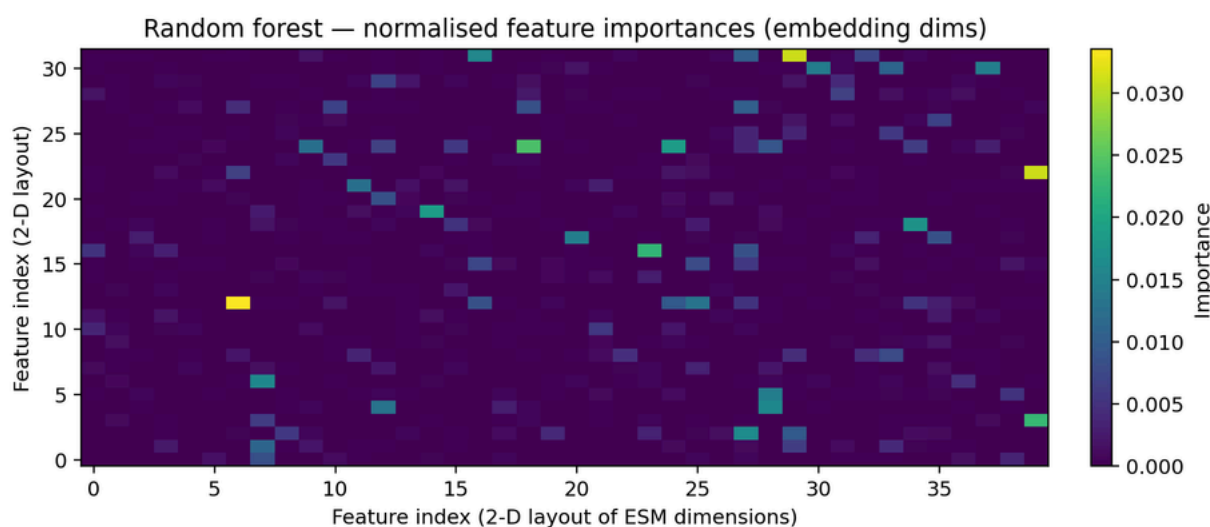


Figure 7.4: Normalized feature importances for the Random Forest classifier. Note that because the input features are 1280-dimensional latent embeddings from the ESM-2 model, individual features lack direct biological interpretability, serving as a limitation of the current transfer-learning approach.

8. TESTING

8.1 Introduction

Testing in a bioinformatics pipeline extends beyond standard software debugging; it requires continuous validation of biological data integrity and mathematical model constraints. The testing strategy for PlasticDeg was formulated to guarantee that data leakage was strictly prevented, structural data transformations (like deduplication) occurred flawlessly, and evaluation metrics were calculated without threshold-tuning biases.

8.2 Testing Tools and Environment

The testing environment was localized within the project's Python conda environment. Data validation tests were programmatic logic checks integrated into the execution scripts (e.g., `integrate_expansion.py`). Machine learning validations utilized scikit-learn's strictly separated fit and predict modules, while the `probe_hard_negatives.py` script served as the primary automated integration testing environment to audit frozen model bounds against completely novel sequences.

8.3 Test Cases

The system underwent rigorous testing across data engineering and machine learning layers. A selection of the critical test cases is documented below:

Test Case ID	Test Objective	Input / Condition	Expected Output	Actual Result & Status
TC-01	Verify BLAST Filter Enforcement	Expanded raw queries against PlasticDB.	Log emits only sequence hits with greater than 90% identity and greater than 90% query coverage.	Filters successfully dropped 17,054 invalid hits. [PASS]
TC-02	Validate Sequence Deduplication	Merging Gold PAZy dataset (162 unique) with Silver expanded set (163 raw).	Overlapping sequences collapse into single records, producing identical ESM-2 hashes.	Final table yielded 189 unique canonical rows. [PASS]
TC-03	Assure Zero Data Leakage	MMseqs2 train and test manifests (split_train_accessions.txt & test variant).	Zero intersecting protein clusters between the two .txt files.	0 overlapping accessions detected. [PASS]
TC-04	Verify Specificity on Easy Negatives	RF Classifier scored against 277 out-of-sample Swiss-Prot test rows.	False Positive Rate evaluates to exactly 0 at frozen threshold (0.8799).	277 True Negatives, 0 False Positives. [PASS]
TC-05	Phase-4 Hard Negative Stress Test	Ingestion of 80 previously unseen EC 3.1.1/3.1.2 esterase and lipase sequences.	No hydrolase sequences exceed the 0.8799 frozen probability boundary.	Max probability score remained safely below threshold (mean = 0.0288). [PASS]

9. CONCLUSION & FUTURE WORK

9.1 Introduction

The rapid accumulation of synthetic plastics in global ecosystems demands scalable and innovative bioremediation solutions. The "PlasticDeg" project successfully addresses a critical bottleneck in this domain by establishing a highly rigorous, computationally honest baseline for discovering novel microbial plastic-degrading enzymes.

By prioritizing data integrity above artificially inflated metrics, the developed pipeline effectively mitigates the common pitfalls of bioinformatics machine learning. The implementation of an MMseqs2-driven homology split successfully eradicated train-test data leakage, while the integration of sequence-level deduplication ensured the model learned from unique biological information rather than redundant accessions. Furthermore, the application of a tier-weighted training strategy effectively balanced highly curated "Gold" PAZy data against homology-propagated "Silver" sequences, yielding a highly precise Random Forest classifier.

The evaluation results unequivocally demonstrate the representational power of the ESM-2 (650M) transformer model. By mapping sequences into a 1280-dimensional latent space, the model achieved perfect precision (1.0) and a zero false-positive rate against general Swiss-Prot negatives at its frozen operating threshold. Crucially, the system passed the Phase-4 hard negative stress test, proving that it does not blindly flag generic EC 3.1.1/3.1.2 hydrolases as polyester-degrading enzymes. Ultimately, PlasticDeg proves that deep transfer learning, when bounded by strict biological evaluation frameworks, is a highly reliable tool for triaging vast microbial genomes and isolating genuine plastic-degrading candidates.

9.2 Future Work

While the PlasticDeg baseline achieves exceptional out-of-sample separability, the intersection of AI and structural biology provides several avenues for future enhancement:

- 1. Integration of 3D Structural Embeddings:** The current pipeline relies exclusively on 1D amino acid sequences via the ESM-2 model. Future iterations should incorporate 3D structural predictions—utilizing tools like AlphaFold2 or ESMFold. Coupling sequence embeddings with structural coordinates (e.g., binding pocket topology) could significantly improve the model's recall and its ability to detect remote functional similarities that fall below standard sequence-identity thresholds.

2. **Wet-Lab Experimental Validation:** Computational predictions inherently serve as a screening mechanism. The most immediate future requirement is to synthesize the highest-confidence predictions—specifically those flagged from uncharacterized metagenomic data—and subject them to in vitro enzymatic assays. Real-world validation of these candidates will provide critical feedback to refine the machine learning baseline.
3. **Enhancing Biological Interpretability (XAI):** The "black box" nature of deep transformer embeddings limits biochemical interpretability. Future work should focus on Explainable AI (XAI) techniques, such as applying attention-mapping algorithms to the transformer heads. This would allow researchers to trace the model's predictions back to specific amino acid residues, potentially highlighting novel catalytic triads or mutational hotspots for directed evolution.
4. **Continuous Metagenomic Pipeline Expansion:** As the plastisphere expands, newly sequenced microbial genomes are constantly deposited into public databases. The current architecture's incremental .npz caching system is designed to scale. Future updates should automate the ingestion of broader metagenomic datasets, utilizing the current frozen model in a semi-supervised learning loop to continuously mine for rare degradation pathways.

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